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Cognitive Mind Meets the Artificial:

*How AI Reshapes Survey Research Through
The Lens of Response Process Theory*

Kevin Linares M.A.

Coauthors:

Rosalina McCoy M.D.,

Alexandria Ratzki-Leewing Ph.D. M.Sc

The Challenge We Face in Health Surveys

...As AI tools...weave their way into survey practice, the need grows to equip researchers with the **theoretical understanding**...¹

Current PROM Limitations²

- Rigid Structure
- Complex medical terminology
- Forced response choices
- Post-hoc scoring

What LLM-PROMs Promise

- Adaptive dialogue
- Plain language clarification
- Open narrative responses
- Real-time Insights



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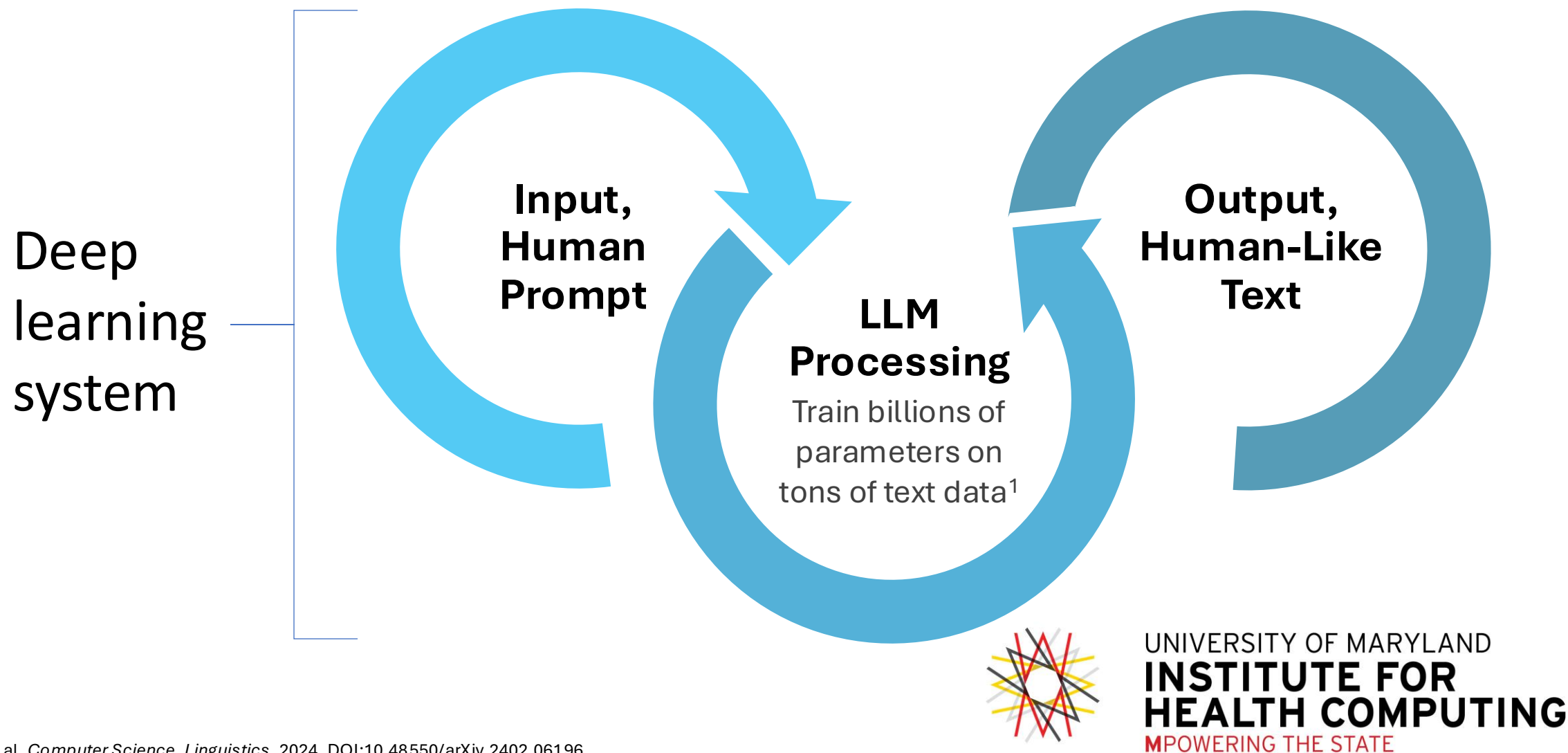
Learning Objectives

- 1 **Explore** LLMs applied in survey research
- 2 **Analyze** LLM-PROMs through Tourangeau's four-stage framework lens
- 3 **Synthesize** implications and practitioner recommendations



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What are Large Language Models?



1. Minaee et al. *Computer Science, Linguistics*, 2024. DOI:10.48550/arXiv.2402.06196

LLM Application in Survey Research

AI Interviewing

Conversational data collection with adaptive questioning.¹⁻³

Response Coding

Automated categorization of open-ended responses.⁴⁻⁶

Questionnaire Design

AI-assisted creation and bias detection.^{7,8}

Translation Support

TRAPD process assistance.⁹

Data Quality Issues

Respondents using AI to generate answers.^{10,11}

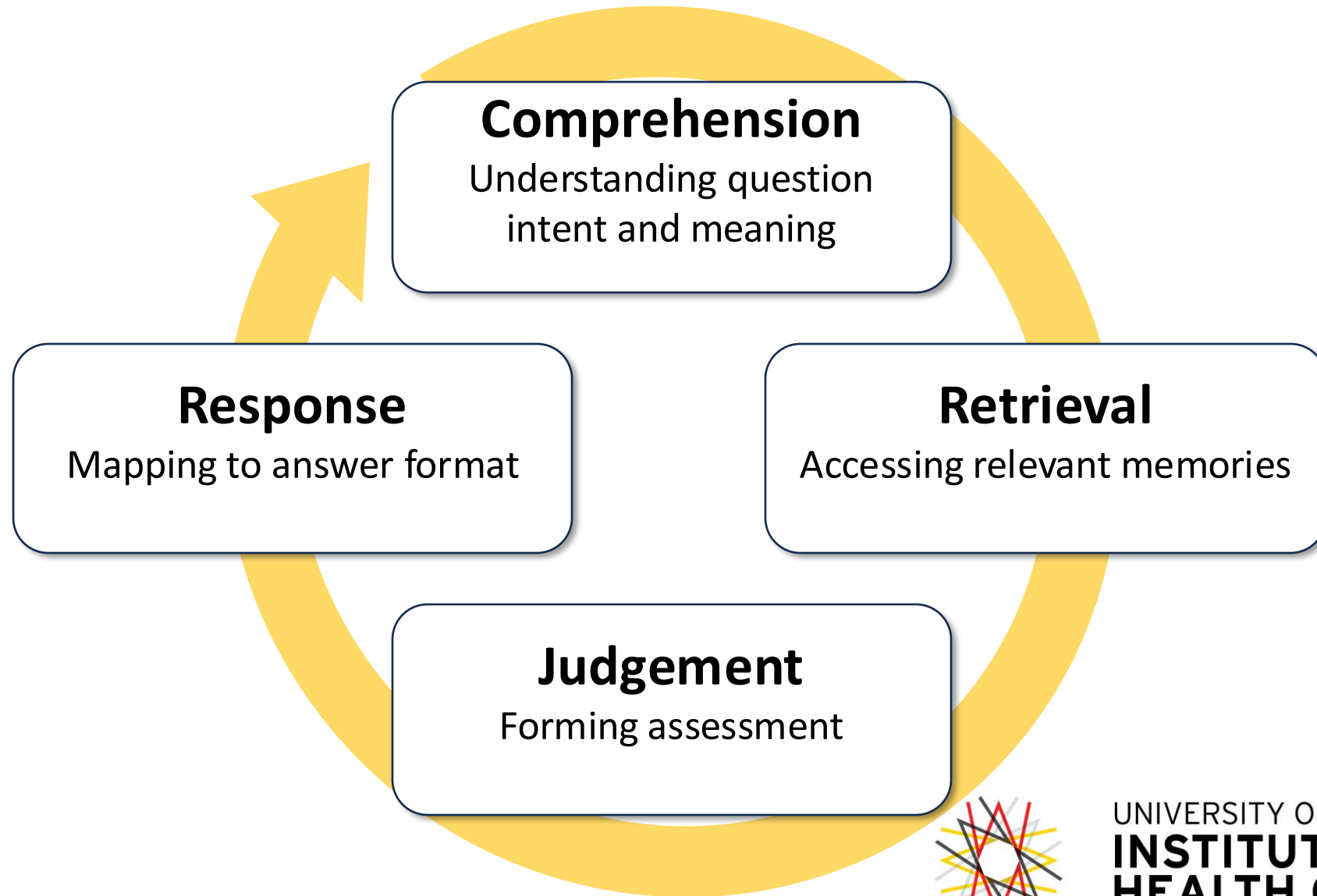
TRAPD (Translation, Review, Adjudication, Pretesting, and Documentation).

1. Jacobsen et al. (2025); 2. Wong et al. (2025); 3. Wuttke et al. (2024); 4. Mellon et al. (2024); 5. Tai et al. (2024); 6. Xiao et al. (2023); 7. NORC (2024); 8. Padgett et al., (2024); 9. Metheney & Yehle (2024); 10. Zhang et al. (2025); 11. Meem et al. (2024)



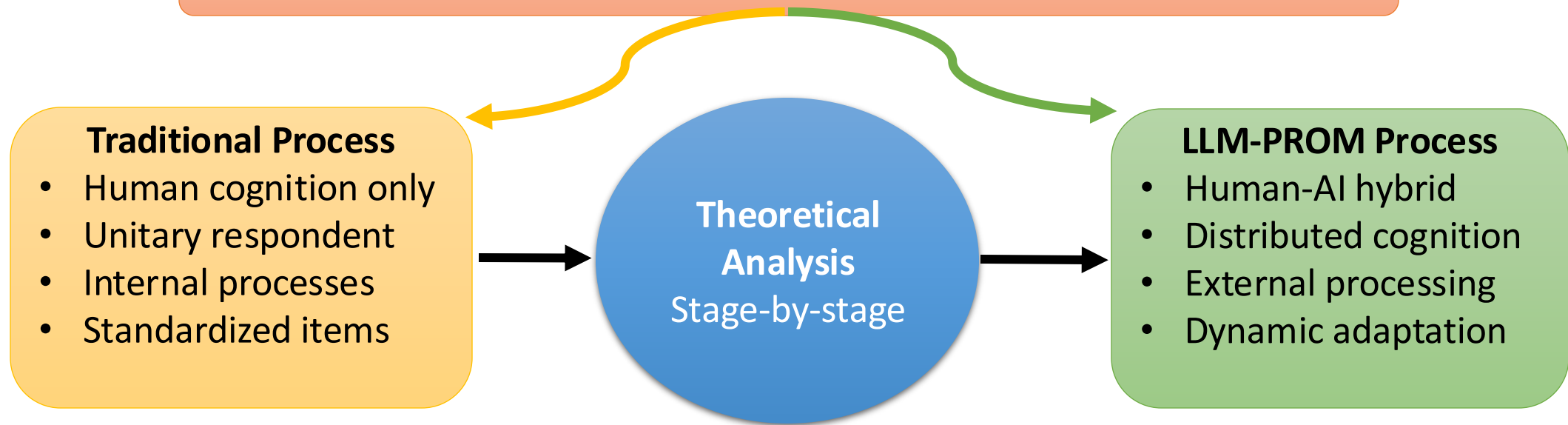
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Tourangeau's (2000) Cognitive Response Process Framework



Theoretical Analysis Approach

Applying Tourangeau's Framework to LLM-PROMs



Goal: Understand how AI fundamentally alters each stage of survey response cognition



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Stage 1: Comprehension Transformation

BENEFITS

Health literacy adaptation

Medical term clarification

Personalized phrasing

CHALLENGES

Loss of standardization¹

Reduced comparability²

AI training bias³

Core Trade-off: Personalization vs. Standardization

1. Conrad & Schober, 2000; 2. Schaeffer & Maynard, 2008; 3. Conrad et al., 2014



Stage 2: Retrieval Transformation

BENEFITS

Systematic symptom reviews

Chronological sequencing

Memory prompting

CHALLENGES

Symptom suggestion¹

False affirmation²

Memory contamination³

Critical Risk: AI suggestions may create false memories or symptom endorsement

1. Fisher & Quigley, 1992; 2. Lee et al., 1999; 3. Conway, 2005



Stage 3: Judgment Transformation

BENEFITS

Reduced coder burden

Rapid processing

Consistent interpretation

CHALLENGES

Loss of patient agency¹

Missing emotional factors²

Context interpretation gaps³

Fundamental Shift: From internal patient evaluation to external algorithmic processing

1. Tourangeau, 2000; 2. Conrad et al., 1998; 3. Brown & Siegler, 1993



Stage 4: Response Transformation

BENEFITS

Rich narrative capture

Reduced cognitive burden

Holistic health dimensions

CHALLENGES

Hybrid expression¹

Systematic response bias²

Measurement equivalence³

Core Issue: no longer measure pure patient expression

1. Suchman, 2007; 2. Lind et al., 2013; 3. Schwarz, 2010

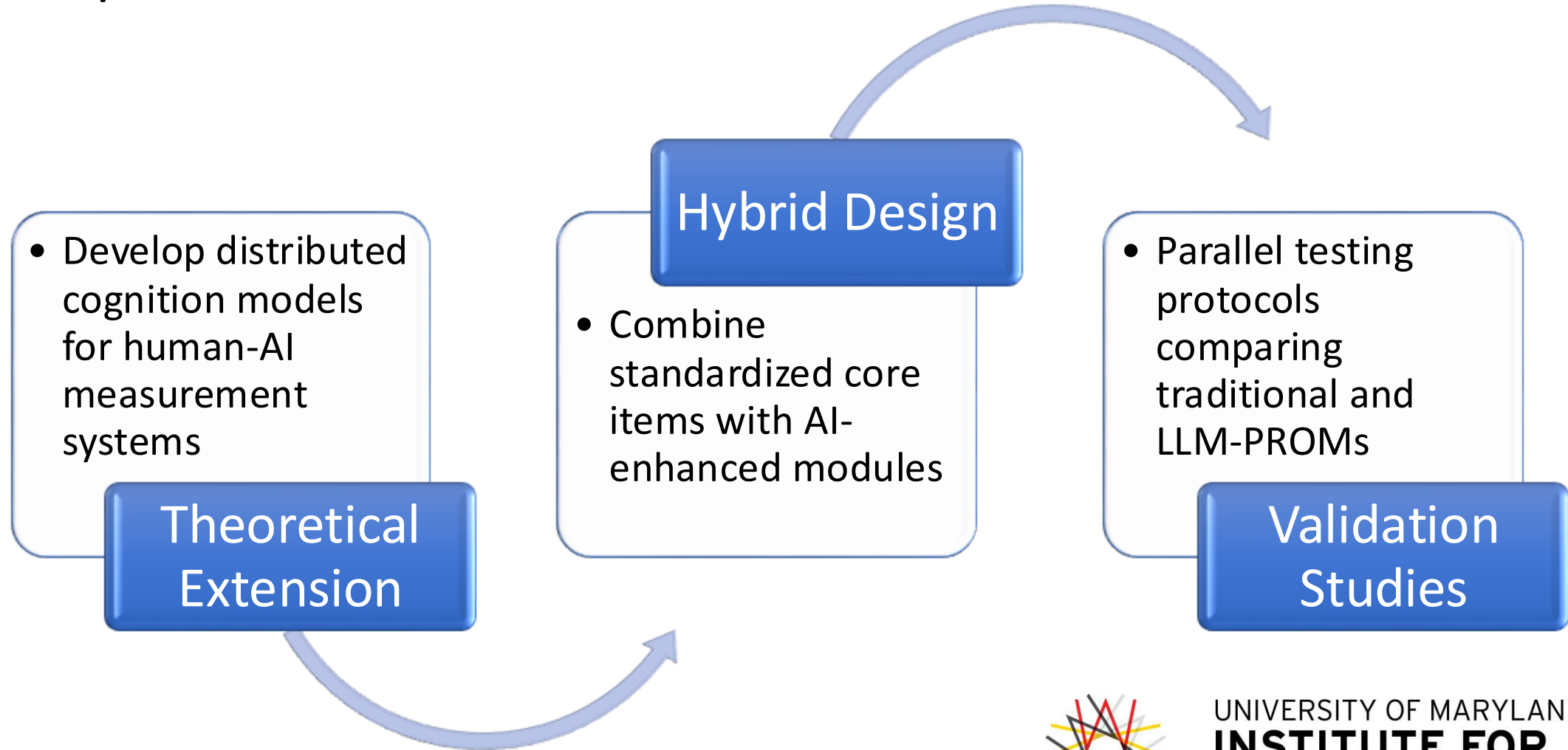


Critical Trade-offs Summary

Stage	What We Gain	What We Lose
Comprehension	Accessibility	Standardization
Retrieval	Memory support	Authentic recall
Judgement	Efficiency	Patient agency
Response	Rich data	Authentic voice



Implementation Framework



Critical Recommendation: Patient Partnership

Amplify, not replace, patient voices

- Patients as co-designers with decision making power
- Testing across diverse communities
- Feedback mechanisms shaping tool evolution
- Transparency about response processing



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Conclusion

LLM-PROMs offer genuine potential but require fundamental rethinking of measurement validity

- ✓ Each cognitive stage fundamentally transformed
- ✓ Real benefits coupled with serious validity threats
- ✓ Patient partnership is non-negotiable for success



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Thank You!

Questions?
Kevin Linares
Klinares@umd.edu

